MILITARY SPECIFICATION

DETONATOR, ELECTRIC, XM65
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification contains requirements not normally required by the drawings and provides for quality assurance provisions for the loading, assembling and packing for one type of initiating device designated as Detonator, Electric, XM65.

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-A-2550 - Ammunition and Special Weapons; General Specification for.
MIL-P-14077 - Polytetrafluoroethylene (Teflon), Molding and Extruding Material.
MIL-I-45208 - Inspection Requirements, General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).

FSC 1375

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MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8798134 - Ammunition Container, Metal for Detonators Electric.
8798135 - Box, Packing, Ammunition for Detonators Electric.
8798136 - Box, Set-up, Packing, Ammunition for Detonators Electric.
9204230 - Detonator, Electric XM65 Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL-9204230 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Detonator.-The detonator shall comply with all requirements specified on Drawing (dwg.) 9204230 and with all requirements specified in applicable specifications.

3.3 Moisture content of lead azide.-The moisture content of the lead azide used in loading the detonator, at the time of loading, shall not exceed 0.50 percent when tested as specified in 4.3.1.

3.4 Moisture content of PETN.-The moisture content of the PETN used in loading the detonator, at the time of loading, shall not exceed 0.50 percent when tested as specified in 4.3.2.

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3.5 Waterproofness.-The detonator shall be waterproof as evidenced by normal functioning after being immersed in water maintained at a temperature of 20 degrees Fahrenheit (F.) plus or minus 5 degree F., for a period of 48 hours, minimum, at a depth of 2 to 3 inches when tested as specified in 4.3.5.

3.6 Functioning

3.6.1 Time.-The detonator shall fire on the first attempt within 10 microseconds when the electrical energy from a .0022 microfarad capacitor, maximum, charged to 300 volts, maximum, is applied to the detonator when tested as specified in 4.3.4.

3.6.2 Output.-The detonator shall fire on the first attempt when the electrical energy from a .0022 microfarad capacitor, maximum, charged to 300 volts, maximum, is applied to the detonator and the detonator shall cause perforation of the lead disc to a diameter of not less than .400 inch when tested as specified in 4.3.5.

3.6.3 Duds.-The detonator shall fire on the first attempt when the electrical energy from a .0022 microfarad capacitor, maximum, charged to 300 volts, maximum, is applied to the detonator when tested as specified in 4.3.6.

3.7 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may

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utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208. The contractor's quality assurance system shall contain a procedure for the reporting of critical defectives found during the course of his inspection and during inspection by the Government. Any critical defect encountered, unless annotated by an asterisk (*) to the right of the characteristic code number (see 6.2), will be reported within 48 hours. A monthly summary report covering those critical defects annotated by an asterisk will be furnished. The agencies to whom the reports will be furnished are identified in paragraph 6.4. The reports shall contain, as a minimum, the following information:

a. Contractor and contract number.
b. Number and revision date of specification.
c. Characteristic code number.
d. Lot or serial number.
e. Number of defectives found.
f. Assignable cause or analysis of occurrence.
g. Corrective action taken or recommended. When the corrective action is beyond the control of the contractor or the scope of the contract, the report shall contain the contractor's recommendation to the Government.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
b. Number of units of product inspected.
c. Results obtained for all inspection performed.

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d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on the PETN, lead azide and all associated ingredients and certificates of compliance on all other material when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of detonators produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot interfix number. Lots shall be submitted for inspection in accordance with Standard MIL-STD-105 (or Standard MIL-STD-1235 when applicable). Each lot shall contain:

a. Parts bearing the same lot interfix number from one manufacturer.

b. Lead azide from not more than one lot.

c. PETN from not more than one lot.

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4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used for defects other than critical, if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Bridge and plug assembly (prior to application of spot charge) (see dwg. 9204231 covering a detail of dwg. 9204230).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Bridge missing	Visual	01001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	01002

4.2.2.2 Bridge and plug assembly (see dwg. 9204231 covering a detail of dwg. 9204230).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Spot charge missing or contains crack, void or separation	Visual	02001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.7)	Visual	02002

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4.2.2.3 Assembly (see dwg. 9204230).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cup split or cracked	Visual	03001 *
Major:	AQL 1.00 percent		
101.	Diameter, max	Gage	03002
102.	Total length, max	Gage	03003
103.	Length excluding pin, min	Gage	03004
104.	Length of pin, min	Gage	03005
105.	Concentricity of pin with diameter of cup	Gage	03006
Minor:	AQL 1.50 percent		
201.	Crimp not full 360 degrees	Visual	03007
202.	Seal missing or incomplete	Visual	03008
203.	Evidence of poor workmanship (see 3.7)	Visual	03009

4.2.2.4 Ammunition, container, metal (see dwg. 8798134).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Contents exposed	Visual	04001
102.	Sealing strip incomplete or wrinkled	Visual	04002
Minor:	AQL 1.00 percent		
201.	Contents move when shaken	Manual	04003
202.	Marking misleading or illegible	Visual	04004

4.2.2.5 Box, set-up, packing (see dwg. 8798136).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			

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Major:	AQL 0.40 percent		
101.	Waterproof seal broken	Visual	05001
102.	Sealing strip improper	Visual	05002
Minor:	AQL 1.00 percent		
201.	Contents move when shaken	Manual	05003
202.	Marking misleading or illegible	Visual	05004

4.2.2.6 Sealed barrier bag (see dwg. 8798136).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Seal improper or incomplete	Visual	06001
102.	Bag punctured, torn or cut	Visual	06002
Minor:	AQL 0.65 percent		
201.	Marking missing, misleading or unidentifiable	Visual	06003

4.2.2.7 Box, packing (see dwg. 8798135).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Strapping missing, broken or loose	Visual-Manual	07001
102.	Contents exposed	Visual	07002
103.	Board broken or split	Visual	07003
Minor:	AQL 1.00 percent		
201.	Contents move when shaken	Manual	07004
202.	Marking misleading or illegible	Visual	07005

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4.2.3 Testing

4.2.3.1 Moisture content of lead azide (see 3.3) - Major defect - Code No. 08001.-The moisture content of the lead azide shall be determined for each lot at the time of loading. A sample of 2 grams shall be selected and tested as specified in 4.3.1. If the moisture content is in excess of the percentage specified and loading has not begun, the lot of lead azide shall be rejected until proper corrective action has been accomplished as verified by repeating the moisture content test specified in 4.3.1. If loading has begun, action on the unloaded lead azide shall be as specified above and all detonators loaded with the questionable lead azide shall be rejected.

4.2.3.2 Moisture content of PETN (see 3.4) - Major defect - Code No. 09001.-The moisture content of the PETN shall be determined for each lot of the time of loading. A sample of 5 grams shall be selected and tested as specified in 4.3.2. If the moisture content is in excess of the percentage specified and loading has not begun, the lot of PETN shall be rejected until proper corrective action has been accomplished as verified by repeating the moisture content test specified in 4.3.2. If loading has begun, action on the unloaded PETN shall be as specified above and all detonators loaded with the questionable PETN shall be rejected.

4.2.3.3 Resistance between pin and cup (see dwg. 9204230) - Major defect - Code No. 03012.-This test shall be performed 100 percent. The test shall be performed as specified in 4.3.3, using test equipment specified in 4.2.4. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot.

4.2.3.4 Functioning and waterproofness.-A sample of 185 detonators shall be selected from each lot and the sample apportioned as specified below for testing in the sequence specified below.

4.2.3.4.1 Functioning time (see 3.6.1) - Major defect - Code No. 03013 and duds (see 3.6.3) - Major defect - Code No. 03014.-A portion of 35 detonators shall be subjected to the functioning time test specified in 4.3.4, using test equipment specified in 4.2.4. The lot shall be rejected if the mean functioning time, ($\bar{X}$) plus

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the product of 2.27 and the standard deviation of the sample (s) exceeds 10 microseconds. Detonators that fail to fire shall not be considered in the required calculation, but will be replaced with detonators from the remaining portion of the original sample. If three duds are encountered while performing this test the lot shall be rejected and testing for acceptance by the Government shall be stopped. Subsequent testing for waterproofness, duds, and output shall not be conducted. The standard deviation shall be calculated using the formula in 6.6 or its equivalent.

4.2.3.4.2 Waterproofness (see 3.5) - Major defect - Code No. 03015, output (see 3.6.2) - Major defect - Code No. 03016 and duds (see 3.6.3) - Major defect - Code No. 03014.-If the lot has not been rejected in 4.2.3.4.1, a portion of 50 detonators shall be tested as specified in 4.3.5, using test equipment specified in 4.2.4. The lot shall be rejected if two or more detonators fail to produce a perforation in the disc of the required size. Detonators that fail to fire shall be replaced with detonators from the remaining portion of the original sample. If a total of three duds has been encountered during the time and output tests combined, the lot shall be rejected and further testing for acceptance by the Government shall be stopped. If a total of three duds has not been encountered but one detonator fails to create a perforation in the lead disc of the required size, the balance of the original sample shall be tested as specified in 4.3.4. The lot shall be rejected and testing for Government acceptance shall be stopped if and when one or more detonator fails to create a perforation in the disc of the required size or when a total of three duds is encountered in all the combined testing.

4.2.3.4.3 Duds (see 3.6.3) - Major defect - Code No. 03014.-If all of the first 50 detonators tested in 4.2.3.4.2 function satisfactorily in the output phase, and the lot has not been rejected for duds; the remaining portion of the original sample shall be subjected to the test for duds specified in 4.3.6. The lot shall be rejected when and if a total of three duds is encountered in all of the combined testing and testing for Government acceptance shall be stopped. If two duds are encountered, a second sample of 370 detonators shall be selected from the lot and subjected to the test specified in 4.3.6. Testing shall be stopped for Government acceptance and the lot shall be rejected if and when the total of duds in all combined testing reaches three.

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4.2.3.5 Heat seal test of sealed bag (see dwg. 8798156) - Major defect - Code No. 06004.-Sampling, acceptance criteria and test procedures for this test shall be as specified in Specification MIL-P-116.

4.2.4 Inspection Equipment.-Index of Inspection Equipment Number IEL-9204230 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of it's accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 Determination of moisture content of lead azide.-A tared dried aluminum dish, 2 inches in diameter shall be covered with a sheet of teflon (see MIL-P-14077), 2 1/2 inches square and sufficiently flat to form a tight cover. An accurately weighed portion of approximately two grams shall be transferred to the moisture dish while the cover and dish are on the balance pan. The dish

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and contents shall be transferred to an oven and dried at 95 degrees C. plus or minus 2 degrees C. for 2 hours. The cover shall be placed in a sulfuric acid desiccator during the drying period. After drying, the dish and contents shall be quickly transferred to the desiccator and allowed to cool to room temperature. The teflon cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to the percentage of moisture in the lead azide as follows:

$$\text{Percentage of moisture} = \frac{(A-B)\ (100)}{A}$$

where:
A = the weight of the sample before drying.
B = the weight of the sample after drying.

4.3.2 Determination of moisture content of PETN.-Approximately 5 grams of PETN shall be placed in a tared glass weighing dish. The dish and its contents shall be weighed, dried at 95 degrees C., plus or minus 2 degrees C., for one hour, cooled in a desiccator and weighed. The loss in weight shall be calculated to the percentage of moisture in the PETN.

4.3.3 Resistance between pin and cup.-The detonators shall be connected to the applicable test equipment specified in 4.2.4 and the electrical resistance measured using not more than 20 microampere of current. Any assembly which fails to comply with the applicable requirement shall be classed defective. The test shall be conducted behind a suitable barricade.

4.3.4 Functioning time.-The detonator shall be assembled to the specified test fixture and connected electrically to the test circuit specified. The energy from a .002 microfarad capacitor, max., charged to 300 volts, max., shall be discharged to fire the detonator. The functioning time shall be measured as the time from when the energy is releaded until functioning of the detonator causes the electronic counter to stop. The time shall be recorded and the appropriate action taken with respect to the duds.

4.3.5 Waterproofness and output.-The detonator shall be submerged in water for the time specified, and at the depth and temperature specified. The detonator shall be removed from the water and within 15 minutes fired by releasing the energy from a .0022

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microfarad capacitor, max., charged to 300 volts, maximum. Observation shall be made for perforation of the lead disc to the required size and for duds.

4.3.6 Duds.-The detonator shall be connected electrically to the test circuit specified and the energy from a .0022 microfarad capacitor, max., charged to 300 volts, maximum shall be discharged to fire the detonator. Observation shall be made for duds.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.-minimum military package.-The detonators shall be preserved and packaged in accordance with dwg. 8798134.

5.1.2 Intermediate packaging.-The detonators shall be packaged in accordance with dwg. 8798136.

5.2 Packing

5.2.1 Level C.-minimum military pack.-The containers shall be packed in accordance with dwg. 8798135.

5.3 Marking

5.3.1 Unit, intermediate and exterior containers shall be marked in accordance with dwgs. 8798134, 8798135 and 8798136.

5.4 Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify title, number and date of this specification.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

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6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designed in the IEL by the following codes:

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection

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equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagramatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above.

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6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Submission of critical defect reports.-For U. S. Army Munitions Command procurements, one copy of each critical defect report will be furnished the cognizant Government inspection agency; the Army Ammunition Procurement and Supply Agency, ATTN: SMUAP-RO; Picatinny Arsenal, ATTN: SMUPA-ND.

6.4.1 Reporting to other services.-For other than U. S. Army Munitions Command procurements, critical defect reporting shall be as prescribed by the Contracting Officer.

6.5 Check test for possible deterioration.-This test shall be performed only when required by the specification covering subsequent use of this detonator in the next assembly or end item. Two hundred (200) detonators shall be selected for this test.

6.5.1 Functioning time and duds.-Forty (40) detonators shall be subjected to the test specified in 4.3.4. Observation shall be made for the functioning time and all duds that may occur shall be recorded. Duds will be replaced by detonators from the 200 selected. If the mean time ($\bar{X}$) plus 2.03 times the standard deviation (s) exceed 10 microseconds, the lot shall be rejected. The standard deviation shall be computed from the formula in 6.6 or its equivalent. Duds shall not enter into the calculation for standard deviation.

6.5.2 Waterproofness, output, and duds.-Fifty (50) detonators shall be subjected to the test specified in 4.3.5. Observation shall be made of the perforation in the disc, and the duds that may occur shall be recorded. Duds shall be replaced by detonators from the 200 selected. If two detonators fail to adequately perforate the lead disc, a retest shall be permitted using the balance of the detonators selected in 6.5. If during the retest one more detonator fails to adequately perforate the lead disc, the retest shall be stopped and the lot rejected. During the retest the number of duds that may occur shall be recorded. If three or more detonators fail to adequately perforate the lead disc during the original test the lot shall be rejected without retest.

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6.5.3 Duds.-If no retest is required in the output phase, the balance of the detonators selected in 6.5 shall be subjected to the test for duds as specified in 4.3.6. The number of duds that may occur will be recorded. If more than two but not more than six duds are found a retest may be permitted using double the number of detonators selected in 6.5. The lot shall be rejected during retest as soon as the total number of duds observed in 6.5.1, 6.5.2 and 6.5.3 is equal to seven. If in the original sample of 200, seven or more duds are observed the lot shall be rejected without retest.

6.6 Calculation of sample standard deviation.-The following formula or equivalent shall be used in calculating the sample standard deviation:

$$s = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n-1}}$$

where:

s = sample standard deviation.
x = numerical value of individual observations.
$\bar{x}$ = average value of the sample.
n = sample size.

6.7 Check test expense.-The contracting officer should arrange to have the contractor reimbursed for any expense incurred in the performance of the check test for possible deterioration.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project No. 1375-A789

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